

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 2: Arithmetic Expressions

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	2 — Arithmetic Expressions
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Paper 3 is the first hard tier: every item is multi-step. Do brackets first, then $\times$ and $\div$, then + and -; never strictly left-to-right. A minus sign before a bracket flips every inner term's sign; distributing means multiplying every term inside; only addition and multiplication may be freely reordered or regrouped — subtraction and division may not. Watch for sign/term-splitting slips, premature left-to-right evaluation, and modelling a situation with the wrong bracketing. Do not write on this paper — mark answers on your answer sheet.

1. Evaluate: $48 - 3 \times (10 - 2 \times 3) + 5$.

(A) 131

(B) 29

(C) 41

(D) 215

2. Which expression has the value 100?

(A) $4 \times 30 - 5$

(B) $4 \times (30 - 5) \div 1$

(C) $(4 \times 30) - 5 \times 1$

(D) $4 + 30 - 5$

3. Written as a sum of signed terms, the expression $50 - 6 \times 4 + 9$ has which terms?

(A) 50, -6, 4, 9

(B) 50, -24, -9

(C) 50, -6×4 , +9 read as 50, 6, 4, 9

(D) 50, -24, 9

4. Which equality is TRUE?

(A) $200 - (60 + 40) = 200 - 60 - 40$

(B) $200 - (60 + 40) = 200 - 60 + 40$

(C) $200 - (60 - 40) = 200 - 60 - 40$

(D) $200 - (60 + 40) = 200 + 60 - 40$

5. Evaluate: $72 \div (4 + 2) \times 3 - 5$.

- (A) 1
(C) 139
- (B) 9
(D) 31

6. Without fully computing, which is greater: 38×25 or $38 \times 24 + 38$?

- (A) They are equal
(C) $38 \times 24 + 38$
- (B) 38×25
(D) Cannot be decided

7. A taxi charges ₹50 to start plus ₹12 for each km. For a 9 km ride, which expression gives the fare in ₹?

- (A) $(50 + 12) \times 9$
(C) $50 \times 9 + 12$
- (B) $50 + 12 \times 9$
(D) $50 + 12 + 9$

8. Which regrouping of $4 \times 17 \times 25$ is easiest mentally, and what is the product?

- (A) $(4 \times 17) \times 25 = 1700$, but harder
(C) $(17 \times 25) \times 4 = 425$
- (B) $(4 \times 25) \times 17 = 1700$
(D) $4 \times (17 \times 25) = 425$

9. Evaluate: $100 - 2 \times (3 + 4 \times (5 - 2))$.

- (A) 94
(C) 238
- (B) -202
(D) 70

10. By the distributive property, 7×96 equals:

- (A) $7 \times 100 - 4$
(C) $7 \times 100 - 7 \times 4$
- (B) $7 \times 90 + 6$
(D) $7 \times 100 + 7 \times 4$

11. Which pair of expressions is EQUAL?

- (A) $30 - 8 - 5$ and $30 - (8 + 5)$
(C) $30 - 8 - 5$ and $30 - (8 - 5)$
- (B) $30 - 8 + 5$ and $30 - (8 + 5)$
(D) $40 \div 4 \div 2$ and $40 \div (4 \div 2)$

12. Which expression equals 250?

- (A) $5 \times 60 - 10$
(C) $(5 \times 60 - 10) \div 1$
- (B) $5 \times 60 - 10 \times 1$
(D) $5 \times (60 - 10) \div 1$

13. A bookshop sells 6 packs of 8 pencils and gives away 5 pencils. Which expression gives the pencils left, and its value?

- (A) $6 \times (8 - 5) = 18$
(C) $6 + 8 - 5 = 9$
(D) $(6 \times 8 - 5) \times 1 = 43$, but model is $6 \times (8 - 5)$
- (B) $6 \times 8 - 5 = 43$

14. Evaluate: $5 \times 6 - 48 \div 6 + 2 \times 7$.

- (A) 52
(C) 44
- (B) 24
(D) 36

15. In the expression $80 - 24 \div 6 \times 2 + 5$, what is the single signed term coming from $24 \div 6 \times 2$?

- (A) -8 (B) -2
(C) -4 (D) $+8$

16. Without computing, which is greater: 53×19 or $52 \times 19 + 19$?

- (A) They are equal (B) 53×19
(C) $52 \times 19 + 19$ (D) Cannot be decided

17. To add $27 + 46 + 73 + 54$ quickly by pairing into round numbers, the best grouping is:

- (A) $(27 + 46) + (73 + 54)$ (B) $(27 + 54) + (46 + 73)$
(C) addition order is fixed, no regrouping allowed
(D) $(27 + 73) + (46 + 54)$

18. Which statement is TRUE?

- (A) $100 \div (10 \div 2) = (100 \div 10) \div 2$
(B) division is associative, so brackets never matter
(C) $100 \div (10 \div 2)$ and $(100 \div 10) \div 2$ have different values
(D) $100 \div 10 \div 2 = 100 \div (10 \div 2)$

19. Evaluate: $(6 + 4 \times 2) \times (10 - 2 \times 3)$.

- (A) 20 (B) 112
(C) 140 (D) 56

20. Using the distributive property, $8 \times 47 + 8 \times 53$ equals:

- (A) 100 (B) 376
(C) 800 (D) 8000

21. Three friends each buy a ₹40 ticket and a ₹15 snack. Which expression gives the total they spend, in ₹?

- (A) $3 \times 40 + 15$ (B) $3 \times (40 + 15)$
(C) $3 + 40 \times 15$ (D) $(3 + 40) \times 15$

22. Which expression equals $100 - 18 - 27$?

- (A) $100 - (18 + 27)$ (B) $100 - (18 - 27)$
(C) $(100 - 18) + 27$ (D) $100 + (18 - 27)$

23. Without full computation, which is greater: $64 + 39$ or $66 + 38$?

- (A) $64 + 39$ (B) They are equal
(C) $66 + 38$ (D) Cannot be decided

24. Evaluate: $90 \div (3 + 2 \times 3) \div 2 + 4$.

- (A) 19 (B) 13
(C) 9 (D) 5

25. A student wrote $60 - (15 - 5) = 60 - 15 - 5 = 40$. What went wrong, and what is the correct value?

(A) Nothing is wrong; the value is 40

(B) The minus before the bracket must flip both inner signs; correct value is $60 - 15 + 5 = 50$

(C) Brackets should be ignored; the value is $60 - 15 - 5 = 40$

(D) The 5 should be doubled; the value is $60 - 15 - 10 = 35$

26. Written as signed terms, the expression $12 \times 3 - 40 \div 5 + 7$ has terms:

(A) 12, 3, 40, 5, 7

(B) 36, -8, -7

(C) 12×3 , $40 \div 5$, 7 i.e. 36, 8, 7 all positive

(D) 36, -8, 7

27. To add $8 + 17 + 12 + 23 + 35$ quickly, which regrouping makes round numbers and what is the total?

(A) $(8 + 17) + (12 + 23) + 35 = 95$, but not round

(B) $(8 + 12) + (17 + 23) + 35 = 95$

(C) $(8 + 23) + (17 + 12) + 35 = 95$

(D) addition cannot be regrouped, total unknown

28. A man earns ₹500 a day for 6 days and spends ₹350 a day for those 6 days. Which expression gives his net savings, in ₹?

(A) $6 \times (500 - 350)$

(B) $6 \times 500 - 350$

(C) $(6 \times 500) - 6 \times 350 + 350$

(D) $6 \times 500 - 350 \times 1$

29. Without computing, which is greater: 25×32 or $25 \times 30 + 25 \times 2$?

(A) 25×32

(B) They are equal

(C) $25 \times 30 + 25 \times 2$

(D) Cannot be decided

30. Evaluate: $200 - 4 \times (12 - 2 \times (3 + 1))$.

(A) 152

(B) 104

(C) 184

(D) 784

31. Evaluate: $7 \times 8 - 6 \times (9 - 4) + 12 \div 4$.

(A) 29

(B) 26

(C) 59

(D) 23

32. Which expression correctly represents 'multiply the difference of 20 and 8 by 5'?

(A) $(20 - 8) \times 5$

(B) $20 - 8 \times 5$

(C) $20 - (8 \times 5)$

(D) $5 \times 20 - 8$

33. Which expression equals $7 \times 100 - 7 \times 2$?

(A) 7×102

(B) $7 \times 100 - 2$

(C) $7 \times 98 - 2$

(D) 7×98

34. Evaluate: $(48 \div 6 + 2) \times 3 - 4 \times 2$.

(A) 38

(B) 22

(C) 14

(D) 30

- 35.** Without computing, which is greater: $81 - 26$ or $83 - 26$?
- (A) $83 - 26$ (B) $81 - 26$
 (C) They are equal (D) Cannot be decided
- 36.** By the distributive property, 12×25 can be split as:
- (A) $12 \times 20 + 5$ (B) $12 \times 20 + 12 \times 5$
 (C) $12 \times 20 \times 5$ (D) 12×25 cannot be split
- 37.** Without computing both fully, which is greater: 99×7 or 98×7 ?
- (A) 98×7 (B) They are equal
 (C) Cannot be decided (D) 99×7
- 38.** Written as signed terms, the expression $100 - 3 \times 6 + 24 \div 8$ has terms:
- (A) 100, -18, -3 (B) 100, -3, 6, 24, 8
 (C) 100, -18, 3 (D) 100, 18, 3
- 39.** Which removal of brackets is CORRECT?
- (A) $120 - (40 - 15 + 5) = 120 - 40 - 15 + 5$ (B) $120 - (40 - 15 + 5) = 120 - 40 + 15 - 5$
 (C) $120 - (40 - 15 + 5) = 120 - 40 + 15 + 5$ (D) $120 - (40 - 15 + 5) = 120 + 40 - 15 + 5$
- 40.** Evaluate: $64 \div 8 \times 2 + 5 \times (7 - 3)$.
- (A) 24 (B) 36
 (C) 21 (D) 84
- 41.** Reena says $24 \div 6 \times 2 = 2$; her teacher says 8. Who is right and why?
- (A) Reena — multiplication is always done before division, so $6 \times 2 = 12$, then $24 \div 12 = 2$
 (B) The teacher — $\times$ and $\div$ are done left to right, so $24 \div 6 = 4$, then $4 \times 2 = 8$
 (C) Reena — you always work right to left
 (D) Both are right; it depends on the order chosen
- 42.** Which expression has the GREATEST value?
- (A) $4 + 2 \times 4 + 2$ (B) $4 \times 2 + 4 \times 2$
 (C) $(4 + 2) \times (4 + 2)$ (D) $4 \times 2 \times 4 + 2$
- 43.** Evaluate: $150 - 6 \times (8 + 4 \div 2) - 10$.
- (A) 80 (B) 68
 (C) 92 (D) -10
- 44.** Which pair of expressions has the SAME value, illustrating the distributive rule?
- (A) $3 \times (5 + 2)$ and $3 \times 5 + 2$ (B) $3 \times (5 + 2)$ and $3 \times 5 + 3 \times 2$
 (C) $3 \times (5 - 2)$ and $3 \times 5 - 2$ (D) $(3 + 5) \times 2$ and $3 + 5 \times 2$
- 45.** A juice stall sells 9 glasses at ₹15 and 7 glasses at ₹25. Which expression gives the total takings, in ₹?

(A) $(9 + 7) \times (15 + 25)$

(B) $9 \times 25 + 7 \times 15$

(C) $9 \times 15 + 7 \times 25$

(D) $9 + 15 \times 7 + 25$

46. Evaluate: $((20 - 8) \div 3 + 1) \times 2 - 3$.

(A) 11

(B) 9

(C) 13

(D) 7

47. Which expression equals 18×99 ?

(A) $18 \times 100 - 1$

(B) $18 \times 100 + 18$

(C) $18 \times 90 + 9$

(D) $18 \times 100 - 18$

48. Evaluate: $5 + 4 \times (6 + 2 \times 3) - 3 \times 4$.

(A) 53

(B) 29

(C) 41

(D) 65

49. A garden has 4 rows of 12 roses and 3 rows of 10 marigolds. Then 8 plants die. Which expression gives the plants left?

(A) $4 \times 12 + 3 \times 10 - 8$

(B) $4 \times 12 + 3 \times 10 - 8$ read as $4 \times (12 + 3) \times 10 - 8$

(C) $(4 \times 12 + 3 \times 10) - 8 \times 7$

(D) $4 + 12 \times 3 + 10 - 8$

50. A student claims $4 + 6 \times 0 = 0$ because 'anything times zero is zero'. The correct value and reason are:

(A) 0, since the whole expression is multiplied by 0

(B) 4, because only $6 \times 0 = 0$ is multiplied first, then $4 + 0 = 4$

(C) 10, since $4 + 6 = 10$ and $\times 0$ is ignored

(D) 6, since $4 + 0 = 4$ then $\times$ nothing

51. Evaluate: $144 \div (2 \times 6) + 3 \times (10 - 2 \times 4)$.

(A) 438

(B) 30

(C) 12

(D) 18

52. Which statement is TRUE about $8 + (6 - 2)$ and $(8 + 6) - 2$?

(A) They differ; $8 + (6 - 2) = 16$ and $(8 + 6) - 2 = 12$

(B) They are equal because subtraction is commutative

(C) They are equal, both 12, because moving the bracket among + and - here doesn't change the value

(D) $(8 + 6) - 2$ is larger because addition comes first

53. Evaluate: $60 - [40 - (15 - 5) \times 2]$.

(A) 80

(B) 20

(C) 40

(D) 100

54. Without computing both, which is greater: 47×53 or 50×50 ?

(A) 47×53

(B) They are equal

(C) Cannot be decided

(D) 50×50

55. Evaluate: $90 - 5 \times 6 + 48 \div (2 + 4) - 2 \times 3$.

(A) 62

(B) 50

(C) 74

(D) 8

56. Which expression correctly represents 'the cost of 3 pens at ₹p each and 2 pencils at ₹q each, then ₹10 discount'?

(A) $3 \times (p + 2 \times q) - 10$

(B) $(3 \times p + 2 \times q - 10)$ read as $3 + p \times 2 + q - 10$

(C) $3 \times p + 2 \times q - 10$

(D) $3 \times p + 2 \times q - 10 \times (3 + 2)$

57. Which statement about $72 \div 8 \times 3 - 5 + 2 \times 4$ is TRUE?

(A) It equals 30, doing $\div$ and $\times$ first (left to right), then $+$ and $-$

(B) It equals 3, working strictly left to right

(C) It equals 24, because $-$ and $+$ come before $\times$ and $\div$

(D) It equals 38, multiplying everything first

58. Evaluate: $84 \div (3 \times 4) + 9 \times 2 - 5$.

(A) 25

(B) 46

(C) 20

(D) 11

59. Which expression equals $(50 + 30) \div 8 + 2 \times 5$?

(A) $80 \div 8 + 10$

(B) $50 + 30 \div 8 + 10$

(C) $80 \div 8 + 2 \times 5$ read as $80 \div (8 + 10)$

(D) $(50 + 30) \div (8 + 2) \times 5$

60. A class of 30 students is split into teams. Each team has 4 players and 1 reserve. How many full teams of 5 can be formed, as an expression and value?

(A) $30 \div 4 + 1 = 8$ (rounded)

(B) $30 \div 4 \times 1 = 7$ (rounded)

(C) $(30 \div 4 + 1) = 6$

(D) $30 \div (4 + 1) = 6$

Solutions — Mathematics (Class 7), Chapter 2: Arithmetic Expressions — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	B	31	A	41	B	51	D
2	B	12	D	22	A	32	A	42	C	52	C
3	D	13	B	23	C	33	D	43	A	53	C
4	A	14	D	24	C	34	B	44	B	54	D
5	D	15	A	25	B	35	A	45	C	55	A
6	A	16	A	26	D	36	B	46	D	56	C
7	B	17	D	27	B	37	D	47	D	57	A
8	B	18	C	28	A	38	C	48	C	58	C
9	D	19	D	29	B	39	B	49	A	59	A
10	C	20	C	30	C	40	B	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Innermost first: $2 \times 3 = 6$, so the bracket is $10 - 6 = 4$. Then $3 \times 4 = 12$. Now $48 - 12 + 5$; do the additions/subtractions left to right: $48 - 12 = 36$, then $36 + 5 = 41$. The 131 answer comes from doing $10 - 2 = 8$ then $\times 3$ inside (wrong order) and mishandling; 29 comes from $48 - 12 - 5$ (treating $+5$ as -5); 215 comes from multiplying $3 \times$ the whole $48-\dots$ chain. Correct value is 41.

2. (B) Take $4 \times (30 - 5) \div 1$: bracket $30 - 5 = 25$, then $4 \times 25 = 100$, then $\div 1 = 100$. The option $4 \times 30 - 5 = 120 - 5 = 115$ (no bracket so $\times 4$ hits only 30). $(4 \times 30) - 5 \times 1 = 120 - 5 = 115$ as well. $4 + 30 - 5 = 29$. Only the first equals 100.

3. (D) A term is a part joined by $+$ or $-$, and $\times$ binds tighter than these. So $6 \times 4 = 24$ is one term, carrying the minus in front: -24 . The terms are 50, -24 , and $+9$. Splitting 6 and 4 apart breaks the product that must stay together as one term, and making the last term -9 wrongly attaches a minus to the $+9$.

4. (A) A minus before a bracket flips the sign of every term inside, so $200 - (60 + 40) = 200 - 60 - 40 = 100$; check directly: $200 - 100 = 100$. ✓. The option $200 - 60 + 40 = 180$ (it kept +40, a sign error). $200 - (60 - 40)$ flips to $200 - 60 + 40 = 180$, not $200 - 60 - 40$. $200 + 60 - 40$ wrongly flips the leading 200 too. Only the first is true.

5. (D) Bracket first: $4 + 2 = 6$. Then $\times$ and $\div$ left to right: $72 \div 6 = 12$, then $12 \times 3 = 36$. Finally $36 - 5 = 31$. Doing $72 \div (6 \times 3) = 72 \div 18 = 4$ then $4 - 5 = -1$ -ish gives the wrong 1; treating $\div$ as last gives 9; multiplying everything gives 139. Correct is 31.

6. (A) By the distributive idea, $38 \times 24 + 38 = 38 \times 24 + 38 \times 1 = 38 \times (24 + 1) = 38 \times 25$. So both name the same number (950). They are equal — no need to multiply out. Picking either side as 'greater' misses that the extra one 38 exactly rebuilds the 25th group.

7. (B) The ₹50 start is paid once; the ₹12 is paid per km for 9 km, i.e. $12 \times 9 = 108$, then add the start: $50 + 108 = 158$. So the fare is $50 + 12 \times 9$. $(50 + 12) \times 9$ wrongly charges the start fee per km. $50 \times 9 + 12$ multiplies the wrong number. $50 + 12 + 9$ just adds raw numbers. Correct is $50 + 12 \times 9$.

8. (B) Multiplication is associative and commutative, so pair the round-number factors: $4 \times 25 = 100$, then $100 \times 17 = 1700$. The other groupings give the same value 1700 but are harder (e.g. $4 \times 17 = 68$ then $\times 25$). The options showing 425 are arithmetic errors ($17 \times 25 = 425$ is correct but then $\times 4$ gives 1700, not 425). Best and correct: $(4 \times 25) \times 17 = 1700$.

9. (D) Innermost bracket: $5 - 2 = 3$. Then $4 \times 3 = 12$, so the outer bracket is $3 + 12 = 15$. Then $2 \times 15 = 30$. Finally $100 - 30 = 70$. Doing $3 + 4 = 7$ then $\times (5 - 2) = 3$ gives $21 \rightarrow 100 - 42 = 58$ -type errors; 94 comes from $100 - 2 \times 3$ only; -202 from multiplying everything left to right. Correct is 70.

10. (C) Write $96 = 100 - 4$, so $7 \times 96 = 7 \times (100 - 4) = 7 \times 100 - 7 \times 4 = 700 - 28 = 672$. Check: $7 \times 96 = 672$. ✓. The option $7 \times 100 - 4 = 696$ forgets to multiply the 4 by 7. $7 \times 90 + 6 = 636$ (the 6 wasn't multiplied). Adding instead of subtracting gives 728. Correct is $7 \times 100 - 7 \times 4$.

11. (A) Removing a bracket after a minus flips inner signs: $30 - (8 + 5) = 30 - 8 - 5 = 17$. ✓. $30 - 8 + 5 = 27 \neq 17$. $30 - (8 - 5) = 30 - 3 = 27 \neq 17$. $40 \div 4 \div 2 = 5$ but $40 \div (4 \div 2) = 40 \div 2 = 20$ — division isn't associative. Only $30 - 8 - 5$ and $30 - (8 + 5)$ are equal.

12. (D) $5 \times (60 - 10) = 5 \times 50 = 250$, then $\div 1$ leaves 250. The other three are all $5 \times 60 - 10 = 300 - 10 = 290$ (the bracket is missing so $\times 5$ hits only 60). So the only one equal to 250 is $5 \times (60 - 10) \div 1$.

13. (B) Total pencils = 6 packs $\times 8 = 48$; give away 5, leaving $48 - 5 = 43$, modelled by $6 \times 8 - 5$. The give-away is once, not per pack, so $6 \times (8 - 5) = 18$ wrongly removes 5 from every pack. $6 + 8 - 5 = 9$ just adds raw numbers. Correct is $6 \times 8 - 5 = 43$.

14. (D) Do $\times$ and $\div$ first: $5 \times 6 = 30$, $48 \div 6 = 8$, $2 \times 7 = 14$. Now $30 - 8 + 14$, left to right: $30 - 8 = 22$, then $22 + 14 = 36$. Treating $+14$ as -14 gives 8; left-to-right from the start gives wrong values like 52 or 44. Correct is 36.

15. (A) The chain $24 \div 6 \times 2$ is one term ($\times$ and $\div$ bind together, left to right): $24 \div 6 = 4$, then $4 \times 2 = 8$. It sits behind a minus sign, so as a signed term it is -8 . Doing $6 \times 2 = 12$ first gives $24 \div 12 = 2$ (wrong order, gives -2); -4 stops at $24 \div 6$; $+8$ forgets the leading minus. The term is -8 .

16. (A) $52 \times 19 + 19 = 52 \times 19 + 1 \times 19 = (52 + 1) \times 19 = 53 \times 19$ by the distributive idea. Both equal 1007. They are equal; the extra one 19 supplies exactly the 53rd group of 19.

17. (D) Addition is commutative and associative, so reorder to make round hundreds: $27 + 73 = 100$ and $46 + 54 = 100$, total 200. The other pairings give $73 + 127$, $81 + 119$ etc., not round. Addition order is definitely not fixed. Best grouping: $(27 + 73) + (46 + 54) = 200$.

18. (C) $100 \div (10 \div 2) = 100 \div 5 = 20$, while $(100 \div 10) \div 2 = 10 \div 2 = 5$. These differ ($20 \neq 5$), so division is NOT associative and the bracket changes the value. The plain $100 \div 10 \div 2$ means $(100 \div 10) \div 2 = 5$, again unequal to 20. The true statement is that the two bracketings have different values.

19. (D) Left bracket: $4 \times 2 = 8$, then $6 + 8 = 14$. Right bracket: $2 \times 3 = 6$, then $10 - 6 = 4$. Product: $14 \times 4 = 56$. Doing $6 + 4 = 10$ then $\times 2$ (wrong order) gives 20 $\rightarrow$ wrong products like 20 or 140. Correct is 56.

20. (C) Factor out the common 8: $8 \times 47 + 8 \times 53 = 8 \times (47 + 53) = 8 \times 100 = 800$. Just adding $47 + 53 = 100$ and forgetting the $\times 8$ gives 100; 376 is 8×47 alone; 8000 multiplies by an extra 10. Correct is 800.

21. (B) Each friend spends $40 + 15 = 55$, and there are 3 of them: $3 \times (40 + 15) = 3 \times 55 = 165$. Equivalently $3 \times 40 + 3 \times 15 = 120 + 45 = 165$. The option $3 \times 40 + 15 = 135$ charges the snack only once. $3 + 40 \times 15$ and $(3 + 40) \times 15$ mismodel entirely. Correct is $3 \times (40 + 15)$.

22. (A) $100 - 18 - 27 = 55$. Factoring the subtracted parts: $100 - 18 - 27 = 100 - (18 + 27) = 100 - 45 = 55$. ✓. $100 - (18 - 27) = 100 - (-9) = 109$. $(100 - 18) + 27 = 109$. $100 + (18 - 27) = 91$. Only $100 - (18 + 27)$ equals 55.

23. (C) Compare by shifting: from $64 + 39$ to $66 + 38$, the first number went up by 2 ($+2$) and the second down by 1 (-1), a net change of $+1$. So $66 + 38$ is greater by 1. Check: $64 + 39 = 103$, $66 + 38 = 104$. ✓. They are not equal. Greater is $66 + 38$.

24. (C) Bracket: $2 \times 3 = 6$, then $3 + 6 = 9$. Now $\times$ and $\div$ left to right: $90 \div 9 = 10$, then $10 \div 2 = 5$. Finally $5 + 4 = 9$. Doing $90 \div (9 \div 2)$ gives 20 (wrong, division not associative) $\rightarrow$ wrong answers; ignoring the $+4$ gives 5. Correct is 9.

25. (B) Inside, $15 - 5 = 10$, so $60 - (15 - 5) = 60 - 10 = 50$. Removing the bracket flips EVERY inner sign: $60 - (15 - 5) = 60 - 15 + 5 = 50$. The student kept the inner -5 as -5 (only flipped the 15), getting 40. So the error is the unflipped 5; correct value 50.

26. (D) $\times$ and $\div$ bind into single terms: $12 \times 3 = 36$ (term $+36$) and $40 \div 5 = 8$, which sits behind a minus so it's -8 ; then $+7$. Terms: 36, -8 , 7. Splitting the products into separate numbers is wrong; making $+7$ into -7 attaches a sign that isn't there; leaving the 8 positive ignores the minus in front. Correct: 36, -8 , 7.

27. (B) Pair to round tens: $8 + 12 = 20$ and $17 + 23 = 40$, plus 35: $20 + 40 + 35 = 95$. The other pairings reach the same 95 (addition is associative) but aren't round (e.g. $8 + 17 = 25$). Addition certainly can be regrouped. Best round grouping gives 95.

28. (A) Each day he saves $500 - 350 = 150$, over 6 days: $6 \times (500 - 350) = 6 \times 150 = 900$. Equivalently $6 \times 500 - 6 \times 350 = 3000 - 2100 = 900$. The option $6 \times 500 - 350 = 2650$ subtracts the spending only once. Correct is $6 \times (500 - 350)$.

29. (B) By distributivity, $25 \times 30 + 25 \times 2 = 25 \times (30 + 2) = 25 \times 32$. Both equal 800. They are equal — the right side is just 25×32 split into two pieces.

30. (C) Innermost: $3 + 1 = 4$. Then $2 \times 4 = 8$, so the outer bracket is $12 - 8 = 4$. Then $4 \times 4 = 16$. Finally $200 - 16 = 184$. Doing $12 - 2 = 10$ then $\times 4 = 40$ (wrong order) gives $200 - 160 = 40$ -type errors; 152 from a different disorder. Correct is 184.

31. (A) Bracket: $9 - 4 = 5$. Products/quotients: $7 \times 8 = 56$, $6 \times 5 = 30$, $12 \div 4 = 3$. Now $56 - 30 + 3$, left to right: $56 - 30 = 26$, then $26 + 3 = 29$. Treating $+3$ as -3 gives 23; ignoring the divide term gives 26. Correct is 29.

32. (A) 'Difference of 20 and 8' is $20 - 8$, and that whole difference is multiplied by 5: $(20 - 8) \times 5 = 12 \times 5 = 60$. Without the bracket, $20 - 8 \times 5 = 20 - 40 = -20$ multiplies only the 8. $20 - (8 \times 5)$ is the same wrong reading. $5 \times 20 - 8 = 92$ doesn't take the difference first. Correct is $(20 - 8) \times 5$.

33. (D) Factor out 7: $7 \times 100 - 7 \times 2 = 7 \times (100 - 2) = 7 \times 98 = 686$. Adding instead gives $7 \times 102 (= 714)$. $7 \times 100 - 2 = 698$ multiplies only the 100 by 7. So the equal one is 7×98 .

34. (B) Inside the bracket do $\div$ first: $48 \div 6 = 8$, then $8 + 2 = 10$. Then $10 \times 3 = 30$. Separately $4 \times 2 = 8$. Finally $30 - 8 = 22$. Doing $48 \div (6 + 2) = 6$ inside gives a wrong bracket; forgetting the -4×2 gives 30. Correct is 22.

35. (A) The amount subtracted (26) is the same in both, but the starting number is larger in $83 - 26$ ($83 > 81$). Starting bigger and taking away the same amount leaves more. Check: $81 - 26 = 55$, $83 - 26 = 57$. Greater is $83 - 26$.

36. (B) Write $25 = 20 + 5$, so $12 \times 25 = 12 \times (20 + 5) = 12 \times 20 + 12 \times 5 = 240 + 60 = 300$. Check: $12 \times 25 = 300$. ✓. The option $12 \times 20 + 5$ forgets to multiply the 5 by 12; multiplying the pieces ($12 \times 20 \times 5$) is wrong. Correct split: $12 \times 20 + 12 \times 5$.

37. (D) Both multiply by 7, but $99 > 98$, so 99 sevens is one extra 7 more than 98 sevens; $99 \times 7 = 98 \times 7 + 7$, hence larger. Check: $99 \times 7 = 693$, $98 \times 7 = 686$. Greater is 99×7 .

38. (C) $\times$ and $\div$ form single terms: $3 \times 6 = 18$ behind a minus $\rightarrow -18$; $24 \div 8 = 3$ with a plus $\rightarrow +3$. Terms: 100, -18 , 3. Making the last -3 wrongly flips the $+$; splitting into single numbers breaks the products; leaving 18 positive ignores its minus sign. Correct: 100, -18 , 3.

39. (B) A minus before the bracket flips EVERY inner sign: -40 stays -40 , -15 becomes $+15$, $+5$ becomes -5 . So $120 - (40 - 15 + 5) = 120 - 40 + 15 - 5 = 90$. Check the bracket value: $40 - 15 + 5 = 30$, and $120 - 30 = 90$. ✓. The other options flip only some signs. Correct: $120 - 40 + 15 - 5$.

40. (B) Bracket: $7 - 3 = 4$. Then $\times$ and $\div$ left to right: $64 \div 8 = 8$, $8 \times 2 = 16$; and $5 \times 4 = 20$. Now $16 + 20 = 36$. Doing $64 \div (8 \times 2) = 4$ (wrong, $\div$ and $\times$ go left to right) gives $4 + 20 = 24$. Correct is 36.

41. (B) Multiplication and division share the same level and are done left to right: $24 \div 6 = 4$, then $4 \times 2 = 8$. So the teacher's 8 is correct. Reena multiplied $6 \times 2 = 12$ first (treating $\times$ as higher than $\div$), giving the wrong 2. There is no 'right to left' rule, and the answer is not a matter of choice.

42. (C) Compute each: $(4 + 2) \times (4 + 2) = 6 \times 6 = 36$; $4 + 2 \times 4 + 2 = 4 + 8 + 2 = 14$; $4 \times 2 + 4 \times 2 = 8 + 8 = 16$; $4 \times 2 \times 4 + 2 = 32 + 2 = 34$. The largest is 36, from $(4 + 2) \times (4 + 2)$.

43. (A) Inside the bracket, $\div$ first: $4 \div 2 = 2$, then $8 + 2 = 10$. Then $6 \times 10 = 60$. Now $150 - 60 - 10 = 80$. Doing $(8 + 4) \div 2 = 6$ inside gives $6 \times 6 = 36 \rightarrow$ wrong; treating -10 as $+10$ gives 100-type errors. Correct is 80.

44. (B) Distributivity gives $3 \times (5 + 2) = 3 \times 5 + 3 \times 2 = 15 + 6 = 21$, and $3 \times (5 + 2) = 3 \times 7 = 21$. ✓. The option $3 \times 5 + 2 = 17$ multiplies only the 5. $3 \times 5 - 2 = 13 \neq 3 \times (5 - 2) = 9$. $(3 + 5) \times 2 = 16 \neq 3 + 5 \times 2 = 13$. Equal pair: $3 \times (5 + 2)$ and $3 \times 5 + 3 \times 2$.

45. (C) Two groups: 9 glasses at ₹15 $\rightarrow 9 \times 15 = 135$, and 7 at ₹25 $\rightarrow 7 \times 25 = 175$; total $135 + 175 = 310$, i.e. $9 \times 15 + 7 \times 25$. $(9 + 7) \times (15 + 25) = 16 \times 40 = 640$ mixes the prices. $9 \times 25 + 7 \times 15$ swaps the prices. Correct is $9 \times 15 + 7 \times 25$.

46. (D) Innermost: $20 - 8 = 12$. Then $12 \div 3 = 4$, and $4 + 1 = 5$. Then $5 \times 2 = 10$. Finally $10 - 3 = 7$. Forgetting the final -3 gives 10; mis-grouping the $\div$ gives other values. Correct is 7.

47. (D) Write $99 = 100 - 1$, so $18 \times 99 = 18 \times (100 - 1) = 18 \times 100 - 18 \times 1 = 1800 - 18 = 1782$. Check: $18 \times 99 = 1782$. ✓. $18 \times 100 - 1 = 1799$ forgets to multiply the 1 by 18. $18 \times 90 + 9 = 1629$. Correct is $18 \times 100 - 18$.

48. (C) Inside: $2 \times 3 = 6$, then $6 + 6 = 12$. Then $4 \times 12 = 48$ and $3 \times 4 = 12$. Now $5 + 48 - 12 = 41$. Doing $(6 + 2) \times 3 = 24$ inside gives a wrong bracket; treating -12 as $+12$ gives 65. Correct is 41.

49. (A) Roses $4 \times 12 = 48$, marigolds $3 \times 10 = 30$, total 78; minus 8 dead = 70, modelled by $4 \times 12 + 3 \times 10 - 8$. The bracketed/ $\times 7$ variants change the meaning, and $4 + 12 \times 3 + 10 - 8$ just chains raw numbers. Correct is $4 \times 12 + 3 \times 10 - 8 (= 70)$.

50. (B) Multiplication binds before addition, so only $6 \times 0 = 0$ is formed first; then $4 + 0 = 4$. The zero multiplies just the 6, not the whole $4 + 6$, so the value is 4, not 0. Treating it as $(4 + 6) \times 0$ is the error. Correct: 4.

51. (D) First bracket: $2 \times 6 = 12$, so $144 \div 12 = 12$. Second bracket: $2 \times 4 = 8$, then $10 - 8 = 2$, and $3 \times 2 = 6$. Now $12 + 6 = 18$. Doing $144 \div 2 \times 6$ without the bracket gives $432 \rightarrow 438$ -type errors. Correct is 18.

52. (C) $8 + (6 - 2) = 8 + 4 = 12$, and $(8 + 6) - 2 = 14 - 2 = 12$. They are equal (both 12). The reason is that here both reduce to $8 + 6 - 2$; it is NOT because subtraction is commutative (it isn't). Neither is larger. True statement: equal, both 12.

53. (C) Innermost: $15 - 5 = 10$, then $\times 2 = 20$. The square bracket is $40 - 20 = 20$. Then $60 - 20 = 40$. A minus-sign slip on the inner term, or doing $40 - 10 \times 2$ as $(40 - 10) \times 2 = 60$, gives wrong answers like 80 or 20. Correct is 40.

54. (D) $47 \times 53 = (50 - 3)(50 + 3) = 50 \times 50 - 9 = 2500 - 9 = 2491$, while $50 \times 50 = 2500$. So 50×50 is greater by 9. (Two numbers with the same sum multiply to a larger product when they are closer together.) Greater is 50×50 .

55. (A) Bracket: $2 + 4 = 6$. Products/quotients: $5 \times 6 = 30$, $48 \div 6 = 8$, $2 \times 3 = 6$. Now $90 - 30 + 8 - 6$, left to right: $90 - 30 = 60$, $60 + 8 = 68$, $68 - 6 = 62$. Sign slips on the $+8$ or -6 give 50 or 74. Correct is 62.

56. (C) Pens cost $3 \times p$, pencils cost $2 \times q$, and a single ₹10 discount is subtracted once: $3 \times p + 2 \times q - 10$. Bracketing as $3 \times (p + 2q)$ mis-groups the items; subtracting $10 \times (3 + 2)$ discounts per item. Correct is $3 \times p + 2 \times q - 10$.

57. (A) Do $\times$ and $\div$ left to right: $72 \div 8 = 9$, $9 \times 3 = 27$; and $2 \times 4 = 8$. Then $27 - 5 + 8$: $27 - 5 = 22$, $22 + 8 = 30$. So it equals 30. Strict left-to-right (treating $\div$, $\times$, $-$, $+$ as one stream) and 'minus before times' are both wrong rules. True: it equals 30.

58. (C) Bracket: $3 \times 4 = 12$, so $84 \div 12 = 7$. Then $9 \times 2 = 18$. Now $7 + 18 - 5$, left to right: $7 + 18 = 25$, then $25 - 5 = 20$. Ignoring the -5 gives 25; doing $84 \div 3 \times 4$ without the bracket gives $112 \rightarrow$ wrong. Correct is 20.

59. (A) Bracket first: $50 + 30 = 80$, and $2 \times 5 = 10$, so the expression is $80 \div 8 + 10 = 10 + 10 = 20$, matching $80 \div 8 + 10$. The option $50 + 30 \div 8 + 10$ drops the bracket (so $\div$ hits only 30). Reading it as $80 \div (8 + 10)$ wrongly brackets the denominator. $(50 + 30) \div (8 + 2) \times 5 = 80 \div 10 \times 5 = 40$, different. Equal one: $80 \div 8 + 10$.

60. (D) Each team needs $4 + 1 = 5$ students, so the number of teams is $30 \div (4 + 1) = 30 \div 5 = 6$. Writing $30 \div 4 + 1$ divides by 4 then adds 1 (wrong grouping); $30 \div 4 \times 1$ ignores the reserve as a teammate. Correct is $30 \div (4 + 1) = 6$.